

LEHIGH VALLEY, PA, USA

WMO: 725170

Lat: 40.650N

Lon: 75.448W

Elev: 119

StdP: 99.90

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.3	(c) -10.9	(d) -21.2	(e) 0.6	(f) -11.9	(g) -18.7	(h) 0.7	(i) -9.2	(j) 11.0	(k) -0.9	(l) 10.0	(m) -2.1	(n) 3.7	(o) 330	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.9	23.4	31.4	22.6	30.0	21.9	24.8	30.4	24.0	29.0	23.3	27.9	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.1	(b) 18.1	(c) 27.3	(d) 22.4	(e) 17.4	(f) 26.7	(g) 21.8	(h) 16.7	(i) 26.0	(j) 75.7	(k) 30.5	(l) 72.5	(m) 29.0	(n) 69.6	(o) 28.0	(p) 28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.3	7.4	-16.4	35.2	3.3	1.8	-18.8	36.5	-20.7	37.5	-22.6	38.5	-25.0	39.8
			-17.1	26.3	3.1	0.9	-19.4	27.0	-21.2	27.5	-22.9	28.1	-25.2	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.5	-0.2	4.5	10.8	16.2	21.2	23.7	22.7	18.9	12.4	6.3	1.4		
	DBStd	9.83	5.60	5.16	5.12	4.64	4.25	3.36	2.85	2.81	3.75	4.40	4.65	4.79		
	HDD10.0	1299	359	287	184	43	3	0	0	0	24	130	269			
	HDD18.3	3019	616	519	430	231	94	12	1	2	38	190	362	524		
	CDD10.0	1823	2	1	13	68	196	336	426	394	266	99	18	3		
	CDD18.3	501	0	0	1	4	29	98	169	138	54	7	0	0		
	CDH23.3	4579	0	1	8	73	332	886	1603	1181	441	53	1	0		
	CDH26.7	1525	0	0	1	17	102	279	610	387	121	8	0	0		
(14) Wind	WSAvg	3.1	3.5	3.6	3.9	3.7	3.1	2.8	2.6	2.3	2.5	2.8	3.0	3.2		
(15) Precipitation	PrecAvg	1119	83	70	87	90	99	97	109	104	111	85	92	89		
	PrecMax	1446	214	138	170	200	270	218	265	239	294	334	246	200		
	PrecMin	758	17	14	23	16	2	23	8	19	19	4	17	16		
	PrecStd	178	46	28	36	43	54	45	61	57	70	57	48	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	17.8	24.2	29.0	31.7	33.5	35.5	34.1	32.6	27.5	21.5	17.9		
		MCWB	15.2	12.8	15.9	18.0	21.5	23.5	24.1	24.1	22.9	20.0	16.5	14.6		
	2%	DB	12.4	13.2	19.2	25.3	29.6	31.4	33.3	32.0	29.8	24.7	19.0	13.9		
		MCWB	10.2	8.7	12.2	16.1	20.5	22.6	23.6	23.3	21.1	18.6	14.9	11.4		
	5%	DB	8.6	9.8	15.8	22.3	27.2	29.7	31.6	30.3	27.7	22.4	16.6	11.0		
		MCWB	6.1	5.8	10.1	14.6	19.1	21.6	22.9	22.3	20.5	17.4	13.0	8.8		
	10%	DB	6.2	7.1	12.8	19.5	24.6	28.2	30.0	28.8	25.9	20.2	14.0	8.6		
		MCWB	3.9	4.1	8.3	12.4	17.6	20.7	22.3	21.5	19.6	15.7	10.6	5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	14.4	16.7	20.2	23.2	25.1	26.0	25.8	24.3	21.8	18.0	15.3		
		MCDB	17.0	17.7	22.1	26.5	29.8	31.1	32.6	31.7	29.8	25.2	20.1	17.3		
	2%	WB	10.6	9.2	13.6	17.7	21.5	23.9	25.0	24.6	23.0	20.1	15.8	12.0		
		MCDB	11.9	12.5	17.7	23.5	27.8	29.2	31.0	29.5	26.9	23.6	18.4	13.7		
	5%	WB	6.7	6.4	11.0	15.6	20.2	22.9	24.1	23.8	22.0	18.4	13.5	9.1		
		MCDB	8.3	9.1	14.8	20.7	25.5	28.0	29.3	28.0	25.6	21.4	15.8	10.7		
	10%	WB	4.1	4.2	8.4	13.7	18.8	21.9	23.4	22.9	20.9	16.5	11.2	6.4		
		MCDB	5.8	7.0	12.4	18.3	23.3	26.4	28.2	26.9	24.4	19.6	13.5	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.2	10.3	11.9	11.6	11.0	11.0	10.9	11.3	11.0	9.9	8.1		
		MCDBR	11.4	13.2	15.1	16.3	15.1	12.8	12.8	12.4	13.5	13.7	13.1	11.0		
	5% WB	MCWBR	9.6	9.7	9.7	9.1	7.2	5.7	5.0	5.2	6.4	7.9	9.3	9.3		
		MCWBR	10.9	12.3	13.7	14.1	13.0	11.2	10.6	10.1	10.4	11.3	11.0	9.9		
(40) Clear-Sky Solar Irradiance	taub	0.305	0.324	0.358	0.387	0.429	0.487	0.487	0.481	0.396	0.361	0.325	0.305			
	taud	2.440	2.367	2.357	2.312	2.225	2.099	2.112	2.116	2.327	2.435	2.496	2.495			
	Ebn at Noon	855	887	892	886	852	800	795	787	840	835	826	828			
	Edh at Noon	79	98	111	125	139	158	155	150	114	92	75	70			
(44) All-Sky Solar Radiation	RadAvg	1.79	2.69	3.58	4.66	5.22	5.68	5.87	5.08	4.20	2.92	2.04	1.51			
	RadStd	0.15	0.31	0.28	0.46	0.60	0.49	0.31	0.32	0.53	0.32	0.19	0.17			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-134	+148	+76		
(47) Regional (64 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+65		

Nomenclature: See separate page